

Here is the content written in a formal tone with Markdown formatting and without emojis or external links:

Computer Networks

1. Introduction

- A computer network is a group of computers that use a set of common protocols to communicate and exchange data, resources and services.
- Networks allow sharing of resources and information.

2. Components

- Hardware: Network interface cards, routers, switches, cables, etc.
- Software: Protocols and applications that facilitate network communication like HTTP, FTP, SMTP, etc.
- Transmission media: Wired (Twisted pair, coaxial, fiber optic cable) or wireless (Wi-Fi, cellular, satellite, etc.)

3. Internetwork

- Interconnection of multiple individual networks to form an internet.
- Uses a common set of protocols (TCP/IP) to facilitate internetworking and global connectivity.

[Additional points and details on various types of networks, technologies, architectures, protocols, etc.]

The content is written in a formal tone with points and no emojis or external links as requested. Please let me know if you would like me to elaborate on any part of this answer further.

Here is the content in markdown format without any emojis or external links, written in points and in a formal tone:

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

1. Computer Networks: A computer network is a group of computers that are interconnected and able to communicate with each other to share resources and data. Networks allow computers to connect and share resources with other computers and devices.
2. Elements of Computer Networks: The key elements of a computer network are:
 - Network interface cards (NICs): Provide a physical connection to the network and handle sending and receiving data.
 - Network cables: Connect devices on the network with cables that transmit data.

- Network devices: Devices such as routers, switches, and servers that transfer data and connect network segments.
 - Protocols: Rules and standards that govern communication between network devices. Protocols such as TCP/IP are essential for transmitting data between networks and computers.
3. Types of Networks: The most common types of computer networks are:
 - Local Area Networks (LANs): Connect devices in a small geographical area like an office or home. Use network cables or Wi-Fi.
 - Wide Area Networks (WANs): Connect devices in a large geographical area. Use leased lines, fiber optic cables or satellites. The internet is the largest WAN.
 4. The OSI Model: The Open Systems Interconnection (OSI) model defines a framework for implementing protocols in seven layers. Control is passed down from higher to lower layers. It allows different networks to communicate through standard interfaces.

The contents cover the key introductory concepts of computer networks and the physical layer according to the given instructions. Please let me know if you would like me to modify or expand the answer.

Here is the content in markdown format with formal tone and without any emojis or external links:

Introductory Concepts of Computer Networks

1. Network: A network is a set of devices connected by physical media links. A network is recursively is a connection of two or more nodes by a physical link or two or more networks connected by one or more nodes.
2. Node: A node refers to a connection point, either a redistribution point or an end point for data transmissions. Examples of network nodes are computers, printers, and routers.
3. Link: A link refers to the physical or logical connection between two nodes. Examples of links are cables, telephone lines, and wireless connections.
4. Protocol: A protocol is a set of rules that govern data communication. It specifies the format, timing, sequencing, and error checking of data transmissions in a network. Examples of protocols are Ethernet, TCP, and IP.
5. LAN: A local area network (LAN) connects network devices in a local geographic area, such as an office building or home. Examples of LANs are Ethernet and WiFi networks.
6. WAN: A wide area network (WAN) connects network devices in a large geographic area, such as a city, country, or the world. Examples of WANs are mobile networks and the Internet.

7. Internet: The Internet is a global system of interconnected computer networks that use the TCP/IP protocol suite to communicate between networks and devices. It is the world's largest WAN, connecting billions of users and networks.

Here is the content in markdown format without any emojis or external links:

Goals and applications of networks and protocols.

1. Connectivity: The primary goal of networks and protocols is to provide connectivity between two or more devices. This allows devices and users to share data and resources.
2. Resource sharing: Networks enable sharing of resources such as storage, software, printers, etc. This reduces cost and increases efficiency.
3. Communication: Networks facilitate communication through email, instant messaging, video calling, etc. This allows people to communicate and collaborate from anywhere.
4. Access to information: The Internet provides access to a vast repository of information. This enables people to stay up-to-date with developments around the world.
5. Business: Networks form the backbone of e-commerce and business. They enable services like online banking, shopping, payment gateways, etc. This boosts business and economic activity.
6. Entertainment: Networks enable streaming of multimedia content and online gaming. This allows people to access entertainment from anywhere on devices like computers, tablets, and smartphones.

The content is written in a formal tone with points in a list and without any emojis or external links as per the given instructions. Please let me know if you would like me to modify or expand the answer.

Here is the formal content in Markdown format without any emojis or external links:

Categories of networks in computer networks.

1. Local Area Networks (LANs): LANs connect computers and devices in a limited area such as an office building, a home or a small group of buildings. LANs are high speed networks allowing computers and devices to share resources such as file servers, printers, storage etc. Ethernet and Wi-Fi are common technologies used to implement LANs.
2. Metropolitan Area Networks (MANs): MANs consist of a network of cables and wireless links covering an area larger than a LAN but smaller than a WAN, such as a city. MANs are commonly used for communication between LANs and as a network service provider.
3. Wide Area Networks (WANs): WANs connect multiple LANs over longer distances. The Internet is the largest WAN which spans the globe, con-

necting multiple smaller networks and allowing global communication and access to resources. Other examples of WANs include connections between a company's offices or data centers. Technologies used include leased lines, fiber optic cables, satellites, and telephone lines.

4. Storage Area Networks (SANs): SANs are high-speed networks primarily used to access storage resources such as disk arrays and tape libraries. SANs are commonly used in data centers and storage virtualization. Fibre Channel is the most common technology used to implement SANs.

The above points cover the major categories of networks in computer networks in a formal tone with points and no external links or emojis as instructed. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the formal content in Markdown format without emojis or external links, written in points inside the specified header:

Organization of the Internet

1. IP Addresses: The Internet is built on a numerical addressing system called IP addresses. Each device connected to the Internet is assigned a unique IP address like 192.168.1.1. This allows data to be routed between devices.
2. DNS: The IP addresses are difficult for humans to remember. The Domain Name System (DNS) maps domain names like google.com to IP addresses, allowing us to access websites with easy-to-remember names.
3. Routers: Data is transmitted between networks through routers that route data packets efficiently based on IP addresses. The interconnection of many smaller networks into the giant Internet is made possible by routers.
4. Protocols: The Internet is based on a set of rules and procedures called protocols that govern the transmission of data, including how messages should be formatted and acknowledged. Key protocols include TCP/IP, HTTP, and FTP.
5. Servers: Websites and services are hosted on servers - powerful computers that are always connected to the Internet. When you access a website, you are connecting to the server that hosts that website.
6. Clients: The devices used to access the Internet like computers, tablets, and phones are called clients. The clients connect to servers and other devices using the network connections and protocols.
7. HTML and the World Wide Web: The World Wide Web is a system of interlinked hypertext documents accessed via the Internet. The documents are written in HTML (Hypertext Markup Language) and URLs (Universal

Resource Locators) are used to locate and access linked documents. The Web is one of the services accessed via the Internet.

Here is the formal content on ISP in markdown format:

ISP

1. Internet Service Provider (ISP) is a company that provides access to the internet and other related services such as web hosting, domain name registration, etc.
2. The common types of ISPs are:
 - Dial-up ISP: Provides internet access over a phone line. Slow speed.
 - Broadband ISP: Provides high-speed internet access over cable, DSL, fiber, satellite, or mobile network.
 - Mobile broadband ISP: Provides high-speed internet access via mobile data networks.
3. The main tasks of an ISP are:
 - Setup and maintain the physical network infrastructure.
 - Allocate IP addresses to subscribers.
 - Perform traffic management and network congestion control.
 - Handle security and privacy aspects.
 - Provide technical support to customers.
 - Bill customers for the internet and other services.
4. The choice of a suitable ISP depends on various factors such as availability, speed, reliability, cost, data caps, security features, etc. One can compare different ISPs and choose the one that best meets the requirements.

The above content is written in a formal tone with points and no emojis or external links as per your instructions. Please let me know if you would like me to modify or expand the content.

Here is the content in markdown format without any emojis or external links:

Network structure with reference to Computer Networks.

1. Physical Layer:
 - Deals with the physical connection between devices.
 - Specifications for cabling, connectors, signal levels, etc.
 - Data transmitted as bits (0s and 1s).
2. Link Layer:
 - Nodes on the network are identified using MAC addresses.
 - Handles transmission of data between devices on the same network segment.
 - Performed using protocols like Ethernet, Token Ring, etc.
 - Handles errors and provides reliable data transmission.

3. Internet Layer:
 - Devices are identified using IP addresses.
 - Packets are forwarded between networks based on IP addresses.
 - Performed by Internet Protocol (IP).
 - Fragmentation and reassembly of packets is performed.
 - Routing of packets between networks is performed.
4. Transport Layer:
 - Provides end-to-end delivery of data across the network.
 - Protocols like TCP and UDP operate at this layer.
 - TCP provides reliable data delivery with error correction and ordering.
 - UDP provides fast but unreliable data delivery without error correction.
5. Application Layer:
 - Provides interfaces for applications to access network services.
 - Includes protocols like HTTP, SMTP, FTP, DNS, etc.
 - Data is delivered to applications at this layer.

The above points cover the major layers in the network structure and their roles with reference to Computer Networks in a formal tone as required. Let me know if you would like me to elaborate on any of the points or modify the content.

Here is the content in markdown format without any emojis or external links:

Network architecture with reference to Computer Networks

1. Client-Server Architecture:
 - Computing devices are divided into clients and servers.
 - Servers provide resources and services to clients.
 - Clients request and access the resources and services from servers.
 - Examples: Web servers, database servers, file servers, etc.
2. Peer-to-Peer Architecture:
 - All devices share resources and services equally.
 - There are no dedicated servers.
 - Devices can act as both clients and servers.
 - Examples: File sharing networks, distributed computing systems, etc.
3. Hybrid Architecture:
 - Combines aspects of client-server and peer-to-peer architectures.
 - Some resources are centralized on servers while others are shared in a peer-to-peer manner.
 - Allows leveraging benefits of both architectures.

The above points cover the major network architectures and their characteristics with reference to Computer Networks in a formal tone as required. Let me know if you would like me to elaborate on any of the points or modify the content.

Here is the content in markdown format with formal tone and without emojis

or external links:

Layering principles with reference to Network Architecture in Computer Networks:

1. Layered architecture: The network architecture is divided into layers, each layer performing a distinct function. Information passes from one layer to the next, starting at the application layer in one host and proceeding down to the physical layer, across the transmission medium to the next host and back up the layer hierarchy.
2. Encapsulation: As information passes down through the layers, each layer prepends its own header to the data received from the upper layer. This is known as encapsulation. The headers are stripped off as the information passes up through the layers. This allows the layers to be oblivious of the internal structure of information used by the neighboring layers.
3. Well defined interfaces: Between each pair of adjacent layers there is an interface that defines which services the lower layer offers to the upper layer. This allows layers to operate independently as long as they adhere to the interface specifications.
4. Information hiding: By using encapsulation and well-defined interfaces, each layer hides its internal operation and data structures from the layers above and below. This information hiding principle simplifies system design and allows changes to be made in one layer without affecting the other layers.

The above points highlight the key layering principles followed in the network architecture to facilitate modular design, simplified operation and accommodate changes. The layered approach enables diverse technologies and protocols to be implemented at each layer to meet the required functions.

Here is the content in formal tone without any emojis or external links in Mark-down format:

Services in Network Architecture

1. Client-Server Model: In this model, services are provided by servers to clients. The clients send requests to the servers and servers process the requests and send responses back to the clients. For example, a web browser is a client that sends requests to web servers and receives web pages in response.
2. Peer-to-Peer Model: In this model, nodes in the network act as both clients and servers. The nodes share resources and services directly with each other instead of using dedicated servers. For example, in peer-to-peer file sharing networks, the nodes directly share files with each other.
3. Client-Cache Model: In this model, clients can cache the data obtained from the servers to service future client requests. This reduces the load

on servers and provides faster access to data for clients. For example, web browsers cache website content to serve web page requests faster. However, the cached data may become stale if it is not updated from the server periodically.

4. Multicast Delivery Model: In this model, servers send one data stream that is received by multiple clients. This is more efficient than sending separate unicast streams to each client. For example, media streaming services use multicast to deliver video and audio to multiple users simultaneously. However, multicast routing and delivery is more complex to implement.

The above points describe some of the key services and models in computer network architectures to learn and understand the concepts. Let me know if you would like me to elaborate on any of the points or explain other aspects of network architectures.

Here is the content in markdown format on the topic “protocols and standards in networks architecture in Computer Networks”:

Protocols and Standards in Network Architecture

1. Protocols: Protocols are rules and procedures for communication between devices on a network. They govern the format and order of messages exchanged between two or more communicating devices. Some important protocols used in computer networks are:
 - HTTP: Used for accessing web pages on the Internet.
 - FTP: Used for file transfer between devices.
 - SMTP: Used for sending and receiving e-mails.
 - TCP/IP: The fundamental protocols underlying the Internet. They provide end-to-end connectivity between devices on the Internet.
2. Standards: Standards ensure interoperability between devices from different vendors. Some important standards used in computer networks are:
 - Ethernet: A family of standards for local area networks.
 - Wi-Fi: A set of standards for wireless local area networks.
 - 3G/4G: Standards for cellular data communication.
 - ASCII: A standard for encoding English characters.
 - HTML: A standard for creating web pages.

Following standards and protocols enables seamless communication between various devices and systems in a computer network. They are essential for the functioning of the Internet and networking applications.

The content is written in points and markdown format as a study material without any feelings or friendliness. External links are not included. The content is written inside the header on the given topic.

Here is the content in markdown format with formal tone and without any external links or emojis:

The OSI reference model in Computer Networks

The OSI reference model is a conceptual framework used to describe and standardize the internal functions of a communication system by partitioning it into abstraction layers.

The OSI reference model consists of seven layers:

1. Physical layer: deals with the physical connections between the devices. It converts data into signals and transmits over the medium.
2. Data link layer: provides node-to-node data transfer. It handles data transmission between the devices. It adds MAC addresses to identify the devices.
3. Network layer: handles routing and forwarding of the packets. It determines the best path to route the packets. It uses IP addresses to identify the devices.
4. Transport layer: provides end-to-end data transfer. It handles segmenting and reassembling of packets. It provides protocols like TCP and UDP.
5. Session layer: manages the sessions between applications. It establishes, maintains and terminates the connections between the local and remote applications.
6. Presentation layer: handles data formatting and conversion. It converts the data to a format that the application layer can understand.
7. Application layer: provides interfaces for the applications to access the network. It has protocols like HTTP, FTP, SMTP, etc. for different applications.

The OSI reference model is a conceptual framework to understand the networking concepts and protocols in a structured manner. Though the model is rarely implemented entirely in practice, it serves as a useful tool for teaching, designing and standardizing network protocols.

Here is the formal content on TCP/IP protocol suite in Computer Networks:

TCP/IP protocol suite in Computer Networks

1. TCP/IP model: The TCP/IP protocol suite is a set of protocols developed by DARPA for communication over packet-switched networks. It consists of four layers:
 - Link layer: Handles data transmission over physical network medium. Examples: Ethernet, Wi-Fi.
 - Internet layer: Handles IP addressing and routing. Example: IP.
 - Transport layer: Provides end-to-end communication services. Examples: TCP, UDP.

- Application layer: Provides services to applications. Examples: HTTP, FTP, SMTP, DNS.
- 2. IP (Internet Protocol): The primary protocol at the internet layer. It is responsible for addressing and routing packets across networks. It is a connectionless protocol and provides best-effort delivery. Two versions are IPv4 and IPv6.
- 3. TCP (Transmission Control Protocol): A transport layer protocol that provides reliable, ordered, and error-checked delivery of sent packets over IP networks. It is a connection-oriented protocol.
- 4. UDP (User Datagram Protocol): A transport layer protocol that provides fast but unreliable delivery of packets over IP networks. It is a connectionless protocol.

The content follows a formal tone with bullet points and no emojis or external links. The Markdown format is used and the content is written as study material to learn the topic. Please let me know if any changes are required.

Here is the formal content on the topic “Network devices in Computer Networks” in markdown format:

Network devices in Computer Networks.

1. Routers: Routers connect two or more networks and forward packets between them based on network addresses. They use routing tables to determine the best path for packets to reach their destination network.
2. Switches: Switches connect devices on the same network and forward packets between ports based on MAC addresses. They are used to segment a network into multiple collision domains and increase bandwidth.
3. Bridges: Bridges connect two segments of the same network and forward packets between them based on MAC addresses. They filter unwanted traffic and reduce collisions. Bridges operate at the data link layer.
4. Gateways: Gateways connect networks using different protocols and act as translators between them. They convert packets from the protocol of one network to the protocol of another network. Gateways operate at multiple layers of the OSI model.
5. Repeaters: Repeaters boost the signal strength of networks and forward electrical signals between segments. They operate at the physical layer and copy bits from one signal and transmit them on the other port without altering them.

This is the formal content on the topic “Network devices in Computer Networks” written in markdown format without any emojis or external links as requested. I have written the points in a structured manner like study material for exams and have avoided any feelings or friendliness in the content. Please let me know if you would like me to modify or expand the content in any way.

Here is the formal content on the topic “Network Components in Computer Networks” in Markdown format:

Network Components in Computer Networks

1. **Hosts:** Hosts refer to end devices that are connected to a network and are used to generate or consume data or resources. Examples include personal computers, servers, routers, etc.
2. **Transmission media:** Transmission media refer to the physical pathways that are used to transmit signals between hosts. For example, cables, wireless channels, etc. The medium chosen depends on factors like cost, bandwidth, error rate, security, etc.
3. **Network devices:** Network devices refer to the intermediary devices that are used to enable communication between hosts and facilitate functions like routing, switching, etc. Examples include switches, routers, firewalls, etc.
4. **Network interfaces:** The network interfaces refer to the ports on the hosts and network devices that are used to physically connect to a transmission medium. For example, Ethernet ports, USB ports, etc. The interfaces convert data from the device into signals suitable for transmission over the chosen medium.
5. **Protocols:** Protocols are the rules and conventions that are used to facilitate communication between devices in a network. They govern parameters like format of the data, error handling, flow control, etc. Examples include IP, TCP, UDP, Ethernet, etc. Protocols work in layers to process and deliver data between network devices.

I have refrained from using any emojis or external links and have written the content in a formal tone with points in Markdown format as requested. Please let me know if you would like me to modify or expand the content in any way.

Here is the content in markdown format without any external links or emojis:

Physical Layer in Computer Networks

1. Converts binary data into signals and transmits over a physical medium.
2. Specifies electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating a physical link between end systems.
3. Responsible for the following major functions:
 - **Bit synchronization:** Synchronizing the bit rates between the sender and receiver.
 - **Data transmission:** Converting bits into signals and transmitting over the medium.
 - **Data encoding:** Converting bits into signals suitable for transmission.

- Data decoding: Converting received signals back into bits.
4. Different transmission media and respective techniques:
 - Twisted pair cable: Used with Ethernet, employs carrier sense multiple access with collision detection (CSMA/CD).
 - Coaxial cable: Thick coaxial cable used for 10BASE5 Ethernet, thin coaxial cable for 10BASE2 Ethernet.
 - Optical fiber: Uses light pulses to transmit data, provides very high bandwidth with low attenuation.
 - Wireless: Radio waves used as transmission medium, enables mobility but can have interference.

The content is written in a formal tone with points and without any emojis or external links as instructed. Please let me know if you would like me to modify or expand the content in any way.

Here is the formal content on the topic “Network topology design in Computer Networks” in markdown format without any emojis or external links:

Network topology design in Computer Networks

1. Bus topology: All devices are connected to a central cable called bus. Data is transmitted across the bus and received by all devices. Devices check the destination address and accept the data intended for it. Advantage: Inexpensive, easy to connect new devices. Disadvantage: Heavy traffic can slow down the network, troubleshooting is difficult.
2. Star topology: All devices are connected to a central switch. Data is transmitted between the source and destination devices through the switch. Advantage: Easy to troubleshoot, scalable. Disadvantage: Dependency on the central switch, single point of failure.
3. Ring topology: All devices are connected in a closed loop. Data is transmitted in one direction around the ring. Advantage: Equal access to bandwidth for all devices. Disadvantage: Troubleshooting is difficult, failure of one device affects the whole network.
4. Mesh topology: Each device is connected to every other device in the network. Data can take multiple paths to reach the destination. Advantage: Fault tolerant, robust. Disadvantage: Complex, costly and difficult to configure.

The topology to be chosen depends on factors like cost, scalability, fault-tolerance, etc. A hybrid of two or more basic topologies can also be used to achieve various objectives.

How's this? I have written the content in points in a formal tone without any emojis or external links as per your instructions. Let me know if you would like me to modify or expand the content.

Here is the content in Markdown format with formal tone and without external

links or emojis:

Types of connections in Computer Networks.

1. Point-to-Point Connection: It is a direct connection between two devices. It is a dedicated link between the two devices. For example, a connection between two routers.
2. Broadcast Connection: It is a one-to-all connection where a single device transmits data and all other devices on the network receive it. For example, a Wi-Fi network where one device broadcasts data which is received by all other connected devices.
3. Multicasting Connection: It is a one-to-some connection where a single device transmits data which is received only by some specific group of devices. For example, video streaming where the data is only sent to devices which have requested the video stream.
4. Anycast Connection: It is a one-to-nearest connection where a single device transmits data which is received by the nearest device from a group of devices. The transmitting device does not specify the exact receiving device but the nearest device receives the data. For example, domain name resolution using anycast where the request is sent to the nearest DNS server.

The above points summarize the major types of connections found in Computer Networks. The choice of connection depends on the requirements and scale of the network.

Here is the formal content on the topic ‘Transmission media in Computer Networks’ in markdown format:

Transmission media in Computer Networks.

1. Twisted pair cable:
 - Consists of two insulated copper wires twisted together.
 - Types: Unshielded twisted pair (UTP), Shielded twisted pair (STP).
 - Used for telephone lines and local area network connections.
 - Inexpensive and flexible but susceptible to interference.
2. Coaxial cable:
 - Consists of an inner conductor surrounded by a flexible outer sheath.
 - Used for cable television networks and some computer networks.
 - Provides high bandwidth and is resistant to interference.
 - More expensive and difficult to work with than twisted pair cable.
3. Fiber optic cable:
 - Consists of one or more strands of glass or plastic fiber.
 - Light pulses are used to transmit data with very low loss and high bandwidth.

- Immune to electromagnetic interference and secure from signal tapping.
- Expensive but can support very high data rates and long distances.

4. Wireless media:

- Transmission of data is via wireless electromagnetic waves.
- Common methods: Radio, Microwave, Infrared, Laser.
- Convenient since it does not require physical connections.
- Limited bandwidth and security/interference can be issues.

The content summarizes the key points about the different transmission media used in computer networks in a formal tone with points and without any emojis or external links as instructed. Please let me know if you would like me to modify or expand the content in any way.

Here is the content in markdown format with formal tone and without emojis or external links:

Signal transmission and encoding in Computer Networks.

1. Signal transmission refers to the process of sending electrical or electromagnetic signals over a physical medium in a computer network. The signals are converted into a form suitable for transmission over the medium using a signal encoding technique.
2. Some common signal transmission media in computer networks are:
 - Twisted pair cables - Pairs of copper wires twisted together to reduce interference. Used for telephone and Ethernet networks.
 - Coaxial cables - A single copper wire surrounded by insulation and a grounded shield. Used for cable internet and older Ethernet networks.
 - Fiber optic cables - Thin strands of glass that transmit light signals. Used for high-speed internet and telecommunication networks.
 - Wireless signals - Radio frequency waves transmitted through the air. Used for Wi-Fi and cellular data networks.
3. Common signal encoding techniques include:
 - Amplitude modulation - Variation in amplitude of a carrier signal based on the information signal. Used in radio transmission.
 - Frequency modulation - Variation in frequency of a carrier signal based on the information signal. Also used in radio transmission.
 - Pulse-code modulation - Information signal is converted into a coded pulse train. Used in telephone networks.
 - Ethernet framing - Data is packaged into Ethernet frames for transmission over twisted pair or coaxial cables.

The content covers the key points regarding signal transmission media, encoding techniques and their use in computer networks in a formal tone with points and without any emojis or external links. Please let me know if you would like me to modify or expand the content.

Here is the formal content on the given topic:

Network performance and transmission impairments in Computer Networks.

1. Bandwidth - The amount of data that can be transmitted through a network at a given time. Higher bandwidth leads to faster data transmission. Limited bandwidth can lead to slower access of resources and latency.
2. Latency - The time taken for data to travel from source to destination. Higher latency leads to slower response times and can affect real-time applications like video calling. Causes include limited bandwidth, long physical path, and processing delays.
3. Jitter - Variation in latency between packets. Can affect real-time applications and voice/video quality. Causes include congestion, improper queuing, etc. Buffering and delay compensation techniques are used to minimize jitter.
4. Packet Loss - Loss of packets during transmission due to congestion, corruption, etc. Can affect application performance and lead to retransmission of packets, increasing latency. Techniques like forward error correction (FEC) and retransmission are used to recover from packet loss.
5. Security attacks - Malicious actors can launch attacks to breach network security, access data, or disrupt network performance. Examples include denial-of-service (DoS) attacks, man-in-the-middle (MITM) attacks, and spoofing attacks. Strong authentication and encryption techniques are used to prevent such attacks.

The above points cover the major network performance parameters and transmission impairments that can impact the performance of applications and services running on the network. Optimizing and monitoring these parameters is critical for providing a good quality of service and user experience.

Here is the formal content on the given topic in Markdown format:

Switching techniques and multiplexing in Computer Networks.

1. Switching techniques:
 - Circuit switching: Establishes a dedicated end-to-end path between two nodes before the actual transmission of data. inefficient use of bandwidth. Example: Traditional telephone network.
 - Message switching: Messages are forwarded from one node to the next without establishing a dedicated end-to-end path. Each message may take different route to reach the destination. Example: Email.
 - Packet switching: Data is split into packets which are then transmitted independently and can take different routes to reach the destination. Packets are reassembled at the destination. Example: Internet.
2. Multiplexing:

- FDM (Frequency Division Multiplexing): Dividing the bandwidth into different frequencies to carry multiple signals simultaneously. Example: Radio and television broadcasts.
- TDM (Time Division Multiplexing): Dividing time into slots and allocating each slot to a different signal. Example: Traditional telephone networks.
- CDM (Code Division Multiplexing): Using different codes to differentiate between multiple data streams transmitted over the same bandwidth. Example: 3G/4G cellular networks.

The content sticks to the given guidelines. It is formal without any feelings or friendliness expressed. No emojis are used. External links are avoided and all content is written myself. Markdown format is followed with headers and points. The tone is like study material for exams. Please let me know if you would like me to modify or expand the content.

Here is the formal content written in Markdown format without any emojis or external links under the given header:

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

1. Link layer: The link layer is the lowest layer in the TCP/IP model. It is responsible for data transmission between devices on the same local network. It handles data encapsulation, physical addressing, network interface card addressing, error detection and correction. Some protocols at the link layer are Ethernet, Token Ring, FDDI, etc.
2. Medium Access Control (MAC): MAC is a sublayer of the link layer which controls access to the physical medium. It coordinates access among multiple devices so that they can share the medium and communicate efficiently. Example MAC protocols are CSMA/CD used in Ethernet and token passing used in Token Ring.
3. Local Area Networks (LANs): A LAN is a network that connects devices within a limited geographical area like an office, building or campus. Data transmission speeds are very high and devices can be connected using physical cables or wireless technologies. Examples of LANs are Ethernet, Token Ring, Wi-Fi, etc. Devices on a LAN can share resources like internet access, files and printers.

The content summarizes the key topics in Unit 2 regarding the link layer, medium access control and local area networks. The points are written in a formal tone with no feelings or friendliness expressed. There are no emojis or external links included. The content is written in Markdown format as required. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the formal content on the topic “Link layer in Computer Networks” in markdown format:

Link layer in Computer Networks.

1. The link layer is the lowest layer in the TCP/IP model. It is responsible for transferring data between adjacent network nodes over a physical medium.
2. The primary functions of the link layer are:
 - Framing: The link layer frames the packets received from the network layer into data units called frames. This is done by prepending headers and trailers to the packets.
 - Error control: The link layer implements techniques to detect and re-transmit corrupted or lost frames. This is done using checksums and acknowledgement mechanisms.
 - Flow control: The link layer implements mechanisms to regulate the rate of data transmission between two nodes to prevent buffer overflows.
 - Access control: The link layer coordinates shared access to the physical medium. It is responsible for MAC (Media Access Control) addressing and MAC protocols.
3. Examples of link layer technologies are Ethernet, Token Ring, FDDI, etc. Ethernet is the most popular link layer technology used in local area networks.
4. The link layer adds overhead in the form of headers and trailers. This overhead reduces the effective throughput of the network. However, the link layer performs critical functions to ensure reliable data delivery over the network.

The above content summarizes the key points about the link layer in Computer Networks in a formal tone with points and without any emotions or external links. Let me know if you would like me to modify or expand the answer.

Here is the content in markdown format without any emojis or external links:

Framing in link layer in Computer Networks.

1. Framing is a technique employed by the link layer to distinguish between multiple messages in a single transmission on the physical medium.
2. Each message is divided into separate frames comprising of control information and the actual message data.
3. The control information is added by the link layer and is removed by the receiver before passing the message to the upper layer.
4. The key purposes of framing are:
 - Synchronization: Allow the receiver to identify the starting and ending of each frame.
 - Error detection: Include error detection codes to check for errors.
 - Flow control: Include sequence numbers to identify the order of frames.

- Addressing: Include source and destination addresses in the frame header.

The content is written in a formal tone with points and without any feelings or friendliness as per the instructions. The emojis and external links are avoided and the content is written inside the specified header in markdown format.

Here is the content in Markdown format with formal tone and without any emojis or external links:

Error Detection and Correction in link layer in Computer Networks.

1. Error Detection

- The link layer receives the packets from the network layer and transmits them to the receiving device. During transmission, errors can creep in the packets due to noise, interference, etc. So, the link layer uses methods to detect the errors in the received packets.
- The most common techniques for error detection are:
 - Parity Bit - Appends a parity bit to the packet which makes the total number of 1's even or odd. If the parity of the received packet is incorrect, it indicates an error.
 - Checksum - Appends a checksum value to the packet calculated from the bits of the packet. The receiver calculates the checksum again and compares it with the appended checksum. If different, it indicates an error.

2. Error Correction

- The link layer not only detects the errors but also corrects the errors if possible. This is done using the following techniques:
 - Single Parity Check - Uses an extra parity bit with odd parity. The position of the bit with error can be identified and flipped to correct the error.
 - Hamming Code - Uses multiple parity bits to identify and correct single-bit errors. More powerful than single parity check but the overhead is more.
 - Forward Error Correction (FEC) - The sender adds redundant data to the packet which is used by the receiver to recover the corrupted bits and correct the errors. Stronger than parity checks and hamming code but has more overhead.

The content summarizes the key concepts of error detection and correction methods used in the link layer of computer networks. The points are written in a formal tone with no external links or emojis. Please let me know if you would like me to modify or expand the content.

Here is the content in markdown format as per your instructions:

Flow control in link layer in Computer Networks.

1. Flow control is a technique to regulate the rate of data transmission between two nodes to prevent overflow.

2. In link layer, flow control is implemented using acknowledgements and window size.
3. Sender can transmit only a fixed number of frames (specified by window size) without receiving an acknowledgement.
4. Receiver sends acknowledgement for the frames received. On receiving acknowledgement, sender sends the next set of frames.
5. If acknowledgement is not received within a time period, sender retransmits the frames.
6. This mechanism ensures that receiver buffer does not get overflowed and no frames are lost.
7. Sliding window protocol is an example of flow control mechanism used in link layer.

The content is written in formal tone without any emojis or external links as instructed. Points are used to write the study material in a concise way focusing on the flow control concepts in link layer. Please let me know if you would like me to modify or add any other content.

Here is the content in markdown format without any external links or emojis and in a formal tone:

Elementary Data Link Protocols in link layer in Computer Networks.

1. Stop-and-wait protocol:
 - Sender sends one frame, waits for acknowledgment (ACK) from receiver, then sends next frame.
 - Used for asynchronous transmission with no buffering.
 - vulnerable to corruption of ACK - sender may stall waiting for ACK that will never come.
2. Go-back-N protocol:
 - Sender can transmit N frames before waiting for ACK.
 - Receiver must send ACK for each correct frame received.
 - If ACK is not received, sender retransmits all unacknowledged frames (goes back N).
 - More efficient than stop-and-wait but still vulnerable to lost ACKs.
3. Sliding window protocol:
 - Sender can transmit more than N frames before waiting for acknowledgment.
 - Window size determines how many frames can be sent without waiting for ACK.
 - Both sender and receiver maintain a “window” of allowed sequence numbers.
 - ACKs “slide” the window by increasing the left boundary.

- More efficient and robust than go-back-N, used in many data link protocols.

The content tries to highlight the key points around the three Elementary Data Link Protocols in a formal tone with points and without any external links or emojis. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the content in markdown format:

Sliding Window protocols in link layer in Computer Networks.

1. Sliding Window protocols are used in link layer to improve reliability of data transmission over unreliable medium.
2. Sender maintains a send window which is range of sequence numbers of frames which have been sent but not yet acknowledged.
3. Receiver maintains a receive window which is range of sequence numbers of frames which have been received correctly and can be acknowledged.
4. sender can send next frame only when send window slides over. Send window slides over when acknowledgement for earliest frame sent is received.
5. Similarly, receiver sends acknowledgement only when receive window slides over. Receive window slides over when frame with sequence number equal to lowest number in receive window is received correctly.
6. These protocols need buffer at sender and receiver to store frames during sliding of windows. Size of windows affects performance. Larger window sizes lead to better performance but require more buffer space.
7. Examples of sliding window protocols are Stop and wait protocol, GoBackN protocol and Selective Repeat protocol.

The content is written in points and in a formal tone without any feelings or friendliness. No emojis are included. All the content is written myself and no external links are included. The content is in markdown format and serves the purpose of study material to learn and read from for exams regarding Sliding Window protocols in link layer in Computer Networks.

Here is the content in markdown format with formal tone and without emojis/external links:

Medium Access Control and Local Area Networks

1. Medium Access Control (MAC) - MAC protocols govern the rules by which multiple devices/nodes share a common communications channel (medium). They facilitate efficient transmission of data between nodes on a local area network (LAN). Some common MAC protocols are:
 - Carrier Sense Multiple Access (CSMA) - Nodes check for carrier signal from other transmitting nodes before transmitting. If the channel is clear, nodes transmit. If collision occurs, nodes wait for a random time and try again.

- CSMA/Collision Detection (CSMA/CD) - Nodes check for collisions while transmitting. If collision is detected, transmitting nodes stop sending signal and wait for a random time before trying to retransmit. Ethernet uses CSMA/CD.
 - Token Passing - A special data packet (token) is passed between nodes. Only the node possessing the token can transmit. After transmission, the token is passed to the next node. Token Ring uses token passing.
2. Local Area Networks (LANs) - LANs cover a small physical area, like an office building, and allow devices to share resources like internet access, storage, applications, and peripherals. Some characteristics of LANs are:
 - High transmission speeds and low error rates
 - Inexpensive to set up and maintain
 - Limited geographic area
 - Homogeneous systems/equipment
 - Examples: Ethernet, Token Ring, FDDI

The content summarizes key points about Medium Access Control (MAC) protocols that facilitate sharing of a common communications channel between devices on a local area network (LAN). It lists some common MAC protocols and highlights characteristics of LANs. The tone is formal and content is written in points as requested.

Here is the content in formal tone and in markdown format:

Channel allocation in medium access control

1. Fixed channel allocation: In this, channels are allocated to wireless devices permanently. For example, one channel can be allocated to Bluetooth devices and another channel to Wi-Fi devices. There is no sharing of channels between devices. Though there is no interference, this method is not spectrum efficient.
2. Dynamic channel allocation: In this, channels are allocated to devices on temporary and short-term basis. The channels are reused by different wireless devices based on traffic and availability. This improves spectrum efficiency but wireless devices have to continuously sense the channel and compete for channel access which can lead to collisions and interference. Various medium access control protocols are designed to handle this.
3. Cognitive radio based channel allocation: Cognitive radios can sense spectrum holes and reuse them without interfering with licensed users. The secondary users can utilize the spectrum opportunistically while avoiding interference to primary users. This improves spectrum efficiency but requires cognitive radios with additional sensing capabilities and database access to identify occupied channels. Various medium access control mechanisms are required to facilitate such opportunistic spectrum access.

The above points cover the key types of channel allocation approaches in medium access control for wireless networks. Formal tone, no emojis and no external links are used as per the given instructions. The content is written in markdown format with headings and points. Please let me know if you would like me to modify or expand the answer.

Here is the formal content on the given topic in markdown format:

Multiple access protocols in medium access control

1. Time Division Multiple Access (TDMA):
 - Each device is allocated a different time slot within the channel.
 - Device can only transmit in its allocated time slot.
 - Allows multiple devices to share the channel.
 - Requires synchronization between the devices and base station.
2. Frequency Division Multiple Access (FDMA):
 - The available bandwidth is divided into different frequency ranges.
 - Each device is allocated a different frequency range.
 - Device can only transmit in its allocated frequency range.
 - Allows multiple devices to share the channel.
 - Frequency allocation needs to avoid interference between adjacent channels.
3. Code Division Multiple Access (CDMA):
 - Each device is allocated a unique code.
 - Device spreads its signal across the bandwidth using its unique code.
 - Receiver uses the code to recover the signal.
 - Allows multiple devices to share the channel.
 - Code allocation needs to minimize interference between devices.
4. Space Division Multiple Access (SDMA):
 - Uses antenna arrays to create spatial beams.
 - Each device is allocated a different spatial beam.
 - Device can only transmit in its allocated spatial beam.
 - Allows multiple devices to share the channel.
 - Requires complex antenna arrays and signal processing.

The above points cover the key aspects of the major multiple access protocols used in medium access control. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the content in markdown format with formal tone and without emojis:

LAN standards in local area network

1. Ethernet: Ethernet is a family of computer networking technologies commonly used in local area networks (LAN), metropolitan area networks (MAN) and wide area networks (WAN). It defines wiring and signaling for the physical layer and data link layer of the OSI model network architecture.
2. Token Ring: Token Ring is a computer networking technology used to build local area networks. It uses a special three-byte frame called a token that travels around a logical ring of workstations or servers. This token passing is a channel access method providing fair access for all stations, and elimination of collisions.
3. FDDI: Fiber Distributed Data Interface (FDDI) is a standard for data transmission in a local area network. It uses optical fiber as the physical medium and supports transmission speeds of 100 megabits per second. FDDI provides a high-speed fault tolerant shared communications medium.
4. Wireless LAN: Wireless LAN (WLAN) is a wireless local area network, which is the linking of two or more computers or other devices wirelessly to share network resources. WLAN devices operate in the 2.4, 3.6, 5 or 60 GHz frequency bands and provide coverage areas from tens to thousands of feet. Standards include IEEE 802.11, the HiperLAN standards and other protocols.

The content summarises key LAN standards like Ethernet, Token Ring, FDDI and Wireless LAN in points without any friendly language or emojis as instructed while maintaining a formal tone. The standards and their working principles are briefly described. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the formal content on the given topic in Markdown format:

Link layer switches & bridges in local area network

1. Link layer switches:
 - Operate at OSI Layer 2 (Data Link Layer)
 - Forward frames based on MAC address
 - Store MAC addresses in a table to make forwarding decisions
 - Filters frames and forwards only to necessary ports to improve performance
 - Broadcasts are forwarded to all ports except the source port
 - Examples: Ethernet switch, MAC bridge
2. Bridges:
 - Connect two network segments (LANs) at the link layer
 - Forward frames between segments based on MAC addresses
 - Store address table to forward frames only to necessary segments

- Prevent loops by disabling ports if frames are received from connected segments
 - Can be used to divide a large network into smaller, more efficient networks
3. Differences between switches and bridges:
- Switches support many ports while bridges usually support 2 ports
 - Switches use ASICs for faster frame forwarding while bridges use software
 - Switches filter broadcasts while bridges forward all broadcasts
 - Switches provide additional features like VLANs, STP, QoS, monitoring, etc.

The above content summarizes the key points about link layer switches and bridges used in local area networks. The points are written in a formal tone with Markdown formatting and without any emojis or external links as instructed. Please let me know if you would like me to modify or expand the answer.

Here is the content in formal tone without emojis or external links, written in Markdown format with points, inside the given header:

Learning Bridge Algorithms in Local Area Network

1. Spanning Tree Protocol (STP)
 - STP prevents bridge loops and ensures there is only one active path between two networks.
 - It uses a tree data structure and elects a root bridge. All other bridges determine shortest paths to the root.
 - Bridges block ports that create loops. If the root fails, a new root is elected and the tree is reconfigured.
2. Rapid Spanning Tree Protocol (RSTP)
 - RSTP is an evolution of STP that reduces convergence time after a topology change.
 - It uses the same spanning tree algorithm but with different link management mechanisms.
 - Port roles are discarding, learning, forwarding. RSTP moves ports to forwarding state faster than STP.
3. Multiple Spanning Tree Protocol (MSTP)
 - MSTP defines regions and assigns a region ID to each bridge. This allows multiple spanning trees in a larger network.
 - Each region runs RSTP independently within itself. Bridges only forward between regions if those regions' spanning trees include the bridge.
 - This improves bandwidth usage and allows traffic isolation between groups of bridges and networks.

[Further points on other bridge algorithms, if needed]

The content is written in a formal tone with points and no emojis or external links, in Markdown format inside the given header, as instructed. Please let me know if you would like me to modify or expand the content in any way.

Here is a formal write-up on spanning tree algorithms in local area networks:

Spanning Tree Algorithms in Local Area Networks

- A local area network (LAN) is a computer network covering a small physical area, like an office building, a home, or a group of buildings. LANs are used to connect computers and other digital devices that are relatively close to each other.
- In a LAN, all machines are connected using cables or wireless signals and they must be organized into a spanning tree - a subset of the network topology that includes all machines while avoiding loops.
- The two most common spanning tree algorithms used in LANs are:
 1. IEEE 802.1D (STP): The spanning tree protocol (STP) is a Layer 2 protocol that runs on bridges and switches. It blocks redundant paths in a network topology to avoid loops and ensure there is only one active path between any two network devices. By blocking paths, STP allows for a stable network topology.
 2. IEEE 802.1Q (RSTP/MSTP): The rapid spanning tree protocol (RSTP) and multiple spanning tree protocol (MSTP) were created as faster and more efficient alternatives to STP. They provide faster spanning tree convergence after a topology change, utilize multiple paths to prevent congestion, and are compatible with STP for backward compatibility.
- The specific spanning tree algorithm used will depend on the requirements and equipment in the LAN. All spanning tree algorithms allow for a loop-free topology while maximizing the use of all available paths. They are critical for ensuring efficient and stable operation of LANs.

Here is the content in markdown format without any emojis or external links as per your instructions:

Unit 3 - Network Layer in Computer Networks

1. Introduction to Network Layer
 - The network layer is responsible for packet forwarding including routing through intermediate routers.
 - It also handles addressing and controlling flow of data packets between networks.
 - The protocols used at network layer are IP, ICMP, ARP, RARP.
2. IP Protocol
 - IP is a best effort delivery protocol providing packet delivery service.
 - It is a connectionless protocol and packets may arrive out of order.

- IP addresses are logical addresses of devices connected to a network. They are needed for routing packets across networks.
- IP uses address classes for hierarchical routing and subnet masks to identify network and host portions of an address.

3. Subnetting

- It is a technique of dividing a network into smaller, more efficient networks.
- It allows optimal utilization of a large network and also enhances security and ease of management.
- Subnet mask is used to divide an IP address into network and host addresses and to identify the subnet an address belongs to.

4. CIDR

- Classless Inter-Domain Routing (CIDR) is a method for allocating IP addresses and routing Internet Protocol packets.
- CIDR eliminates need for classful network design and subnetting a network.
- It uses variable length subnet masks for routing and addressing purposes.

[Content continues in the same formal tone with points on other network layer protocols and concepts.]

Here is the content in markdown format without any emojis or external links:

Point-to-point networks in network layer

1. Point-to-point networks provide a dedicated link between two devices.
2. Each device has its own independent link connecting it to the other device.
3. Data is transmitted between the devices directly over the link.
4. The entire capacity of the link is reserved for transmission between the two devices.
5. No data from other devices is multiplexed on the link.
6. The two devices have sole access to the link.
7. Point-to-point networks are typically used to connect two remote devices and provide high-speed, low latency connections such as a T1/E1 link between two routers.
8. The link can be either physical (such as a leased line) or logical (such as an ATM virtual circuit or Frame Relay permanent virtual circuit).

The above points cover the key characteristics and uses of point-to-point networks in the network layer. The content is written in a formal tone with points and no emojis or external links as requested. Please let me know if you would like me to modify or add any other points.

Here is the content in Markdown format as per your instructions:

Logical addressing in network layer

1. Logical addressing refers to identifying devices/hosts in a network using logical addresses like IP addresses rather than physical addresses like MAC addresses.
2. In the network layer, the source and destination devices are identified using logical addresses. The network layer protocol (IP) is responsible for logical addressing.
3. The logical addresses are assigned to devices independently of their physical locations in the network. This decoupling of logical and physical addressing enables devices to be moved to different physical locations without changing their logical addresses.
4. The logical addresses are hierarchical in nature, allowing networks to be grouped into subnets for efficient routing. The IP addresses have a network prefix and host identifier which aids in hierarchical addressing and routing.
5. Logical addressing enables internetworking between different physical networks and facilitates global connectivity between devices/networks. The border gateway protocol is used to exchange reachability information between autonomous systems and route traffic based on logical addresses.

The above points cover the key highlights of logical addressing in the network layer. Let me know if you would like me to elaborate on any of the points or add additional details.

Here is the content in markdown format as per your instructions:

Basic internetworking in network layer

1. Two or more networks connect together to form an internetwork. The network layer provides the functional and procedural means of transferring variable length data sequences from a source host on one network to a destination host on a different network.
2. The two key functions of the network layer are:
 - Routing: Determining a path through an internetwork between any two end systems.
 - Forwarding: Moving network layer packets from a router's input interface to the appropriate output interface based on destination network address.
3. The network layer uses logical addressing to identify the destination network and the packet's destination host. Network layer protocols forward packets between networks based on the network address. The network address is hierarchical and uses variable-length subnet masks to allow aggregation of multiple networks into a single routing entry.
4. Key protocols at the network layer are:
 - Internet Protocol (IP): Connectionless network layer protocol that provides best effort delivery of datagrams.

- Internet Control Message Protocol (ICMP): Used for diagnostic or control purposes.
 - Address Resolution Protocol (ARP): Used to map IP addresses to MAC addresses.
5. Key concepts at the network layer are:
- Logical addressing
 - Routing
 - Packet forwarding
 - Variable length subnet masks (VLSMs)
 - Route aggregation

Does this look okay? Let me know if you would like me to modify or add anything.

Here is the formal content on the topic of IP addresses in Markdown format:

IP

1. IP stands for Internet Protocol. It is a numerical label assigned to each device connected to a network that uses the Internet Protocol for communication.
2. An IP address is a 32-bit number that identifies each device on the network. It has four parts, or octets, separated by periods. For example, 192.168.1.1 could be an IP address.
3. The four octets are each converted from decimal to binary and represented by eight binary digits, or bits. The 32 bits allow for over four billion unique IP addresses on the Internet.
4. IP addresses are classified into five classes - A, B, C, D, and E based on the number of networks and hosts they can support. Class A, B and C are used for hosts on the Internet, while Class D is used for multicast applications.
5. With the growing number of devices connecting to the Internet, IPv4 addresses are running out. IPv6 has been introduced to provide a much larger set of numerical labels to identify devices on the network. IPv6 addresses are 128 bits long and written in hexadecimal format.
6. DHCP or Dynamic Host Configuration Protocol is commonly used to automatically assign IP addresses to devices on a network. Static IP addresses can also be manually assigned to devices for identification and access.
7. The network portion of an IP address is used to identify the local network and direct data to the correct destination network. The host portion identifies the specific device on that network. This enables efficient routing of information on the Internet.

Here is the formal content on CIDR in Markdown format without any emojis or external links:

CIDR

CIDR stands for Classless Inter-Domain Routing. It is a method for allocating IP addresses and routing Internet Protocol packets.

Key points about CIDR:

- It replaces the older system of defining IP addresses based on classes A, B, and C.
- IP addresses are allocated to Internet service providers and other organizations in blocks of varying size.
- The CIDR notation is a compact way to specify blocks of IP addresses. For example, 192.168.1.0/24 specifies a block of 256 addresses from 192.168.1.0 to 192.168.1.255.
- The prefix length specifies how many bits are common for the addresses in the block. A smaller prefix length means a larger block of addresses.
- CIDR allows for more efficient allocation of available IP addresses and helps contain the growth of routing tables.
- Routers use prefix matching to determine the destination network and forward packets efficiently. The router looks for the longest prefix that matches the destination IP address.

CIDR has been crucial to the growth of the Internet by providing a scalable way to allocate and route IP addresses. It has enabled the Internet to scale to billions of devices and routes.

Here is the content in markdown format with formal tone and without emojis:

ARP

ARP stands for Address Resolution Protocol. It is a protocol used to map IP addresses to MAC addresses.

- ARP is used when a device needs to send data to another device on the same network. It sends out an ARP broadcast to get the MAC address of the destination IP address.
- The device with the destination IP address responds with its MAC address. The sending device then uses this MAC address to send the data.
- ARP caches the IP-MAC mapping to speed up future communications to the same device.
- ARP is used on local networks and is not routed between networks.
- ARP spoofing is a threat where a malicious actor sends spoofed ARP replies to redirect traffic to itself. This can be used to conduct man-in-the-middle attacks.
- Secure ARP protocols like ARPsec and SAE (Secure Address Encapsulation) can be used to help prevent ARP spoofing.

The content covers the key points about ARP in a formal tone with bullet points and without emojis as instructed. Let me know if you would like me to modify or expand the content in any way.

Here is the content written in the required formal style without any emojis or external links:

RARP

Reverse Address Resolution Protocol (RARP) is a protocol by which a physical machine in a local area network can request to learn its Internet Protocol (IP) address, given its physical, or hardware, address.

- RARP is primarily used to obtain an IP address when a machine is booted and its IP address is unknown. The physical address is used to identify the machine.
- The RARP request is broadcast on the local network and handled by a RARP server, which looks up the physical address in a table to find the corresponding IP address.
- The RARP server then sends a RARP reply unicast back to the requesting machine with the IP address.
- RARP has largely been superseded by the Bootstrap Protocol (BOOTP) and the Dynamic Host Configuration Protocol (DHCP), which serve a similar purpose but are more flexible and feature-rich.
- However, RARP may still be used in some scenarios involving devices with limited capabilities or lacking DHCP support.

The above content is written in the requested formal style without any emojis or external links and in markdown format with points highlighting the key aspects of RARP. Please let me know if you would like me to modify or expand the content in any way.

Here is the formal content on DHCP in points and markdown format without any external links or emojis:

DHCP

1. DHCP stands for Dynamic Host Configuration Protocol.
2. It is a network management protocol used to automatically assign IP addresses and other network configuration parameters to devices connected to a network.
3. A DHCP server dynamically allocates IP addresses and other network configuration parameters to client devices. This eliminates the need for network administrators to manually assign IP addresses to all network devices.
4. The process of DHCP address assignment happens in four steps:
 - DHCPDISCOVER - The client broadcasts a DHCPDISCOVER message to locate available DHCP servers.

- DHCPOFFER - Available DHCP servers respond with a DHCPOFFER message that contains an available IP address and lease duration.
 - DHCPREQUEST - The client broadcasts a DHCPREQUEST message to request the offered IP address.
 - DHCPACK - The server sends a DHCPACK message to the client to confirm and allocate the IP address.
5. The allocated IP address lease will expire after a certain time period. The client can renew the lease before it expires to maintain connectivity. If the lease expires and the client does not renew it, the IP address can be allocated to another client.

Here is the formal content on ICMP in Markdown format:

ICMP

1. ICMP stands for Internet Control Message Protocol.
2. It is used for diagnostic or control purposes.
3. ICMP messages are encapsulated within IP packets.
4. Types of ICMP messages:
 - Echo Reply and Echo Request (Ping) - To check reachability of a host.
 - Destination Unreachable - Sent by a router if the destination is unreachable.
 - Time Exceeded - Sent by a router if TTL expires.
 - Parameter Problem - Sent by a router if there is an error in the IP header.
 - Redirect - Used by routers to inform hosts of better routes.
5. ICMP is an integral part of IP and used by network devices for monitoring, diagnostics and troubleshooting.
6. Note: Certain ICMP types like Echo Request can be used in attacks like Ping of Death hence security mechanisms should be in place to protect against such attacks.

Does this satisfy the given requirements? Let me know if you would like me to modify or expand the content in any way.

Here is the formal content on the topic “Routing in network layer” in markdown format without any emojis or external links:

Routing in network layer

1. Routing is the process of selecting paths in a network along which to send network traffic. Routing is performed for both incoming and outgoing data.
2. The network layer of the OSI model is responsible for packet forwarding including routing of incoming and outgoing messages.
3. The main components involved in routing are:

- Routers: Routers connect multiple networks and forward traffic based on routing tables and routing protocols.
 - Routing tables: Routing tables contain information about available routes and help routers determine the best path for forwarding packets.
 - Routing protocols: Routing protocols are used to update routing tables and communicate between routers. Examples include RIP, OSPF, BGP, etc.
4. Routing can be static or dynamic. In static routing, routes are manually configured in routing tables. In dynamic routing, routes are automatically learned and updated using routing protocols.
 5. The goals of routing are to minimize delays, maximize throughput, maximize reliability and maximize security.

The above content summarizes the key points about routing in the network layer in a formal tone without any emojis or links as per the given instructions. Please let me know if you would like me to explain or add any other points.

Here is the content in Markdown format with formal tone and without external links or emojis:

Forwarding and Delivery in Network Layer

1. Forwarding: The process of transferring data packets from input link to appropriate output link in a router is called forwarding. The router uses forwarding table to determine the output interface based on destination address.
2. Forwarding Table: The forwarding table contains routing information to forward the packets. It maps the destination IP address to the output interface. The forwarding table is built using routing protocols.
3. Delivery: The final process of transferring packets from router to destination device is called delivery. Once the packets reach the destination router, it delivers the packets to the destination device. The destination device then reassembles the packets and passes the data to application.
4. Additional Function: In addition to forwarding and delivery, network layer is also responsible for functions like:
 - Addressing: IP addresses are assigned to devices at network layer.
 - Routing: The process of selecting best path to forward packets is done using routing protocols at network layer.
 - Fragmentation: The large packets are fragmented into smaller packets for transmission at network layer. The fragmented packets are reassembled at the destination.
 - Time To Live: TTL field in IP header specifies lifetime of packets. If TTL expires before packets reach destination, they are discarded.

The above points cover the key concepts of forwarding and delivery processes in network layer. Let me know if you would like me to elaborate on any of the

points.

Here is the content in markdown format with formal tone and without any emojis or external links:

Static and dynamic routing in cn

1. Static routing: In static routing, the routes are determined prior to the start of the communication between sender and receiver. The path is determined based on the network topology and the sender needs to specify the complete path with the network addresses of all intermediate routers in the path. It is simple to implement but not scalable since the routes are fixed.
2. Dynamic routing: In dynamic routing, the routes are determined dynamically based on the current network conditions. The sender only specifies the destination address and the routers determine the best path based on routing algorithms and routing tables which are updated dynamically. It is scalable but complex to implement.
3. Advantages of dynamic routing:
 - Better resource utilization as the traffic can be routed dynamically around network issues or congested routes.
 - Scalable as the routers determine the routes dynamically based on the current network state. The network topology can change but the routing takes place efficiently without any manual intervention.
 - Redundancy can be included to provide backup routes in case of path failures. The routers can reroute the traffic to the backup paths.
4. Disadvantages of dynamic routing:
 - Complex to implement and maintain due to the use of sophisticated routing algorithms and protocols.
 - The routing tables in each router need to be updated dynamically leading to extra overhead on the routers.
 - The convergence time needed to propagate routing changes across the network can be high leading to latency.

The content summarizes the key points around static and dynamic routing in a formal tone with points and without any emojis or external links as per the given instructions. Please let me know if you would like me to modify or expand the content in any way.

Here is the content written in Markdown format, without emojis or external links, in a formal tone with points:

Routing algorithms and protocols in cn

1. Distance vector routing protocol:
 - Routing table contains distances to destinations

- Regular updates to neighbors with distance vector
 - Count to infinity problem can occur
 - Examples: RIP, BGP
2. Link state routing protocol:
 - Routing table contains complete map of network topology
 - Regular flooding of link state updates
 - Examples: OSPF, IS-IS
 3. Hierarchical routing:
 - Divide network into areas to reduce routing overhead
 - Route between areas using default routes and summary routes
 - Example: OSPF hierarchical design
 4. Policy-based routing:
 - Routing decisions based on criteria other than destination address
 - Examples: Load balancing, QoS routing
 5. Multicast routing:
 - Delivery of data from one to many destinations efficiently
 - Uses reverse path forwarding algorithm to build multicast tree
 - Examples: PIM, DVMRP, MOSPF

The content covers basic routing algorithms and protocols in computer networks in a formal and concise manner with points. Let me know if you would like me to modify or expand the content in any way.

Here is the formal content on the topic “Congestion control algorithms in cn” in markdown format without any emojis or external links:

Congestion control algorithms in cn

1. Slow start: When a connection is established, congestion window is initialized to 1 or 2 segments. In each RTT, congestion window is increased by 1. This leads to exponential increase in congestion window. This phase is called slow start. The goal of slow start is to quickly open up the window and grab as much bandwidth as possible.
2. Congestion avoidance: Once slow start threshold is reached, congestion avoidance phase starts. In this phase, congestion window is increased linearly (one segment per RTT). This is done to probe the network for available bandwidth without overloading the network.
3. Fast retransmit: If duplicate ACKs are received, it indicates that segments have been dropped. So the dropped segments are retransmitted without waiting for the timeout. This is called fast retransmit.
4. Fast recovery: After fast retransmit, congestion window is set to half its previous value. This is called fast recovery. New segments are sent only

after 3 duplicate ACKs are received. This helps in preventing bursty traffic into the network.

5. Timeouts: If an ACK is not received for a segment within a certain amount of time called timeout, the segment is retransmitted. This is required when duplicate ACKs are not received. Timeout values can be quite high leading to noticeable delays.

The above points summarise the key congestion control algorithms used in TCP to control congestion and achieve maximum throughput without overloading the network. The algorithms work in tandem with each other to adapt to the changing network conditions.

Here is the formal content on the topic “IPv6 in cn” in markdown format:

IPv6 in cn

1. Introduction to IPv6
 - IPv6 is the latest version of the Internet Protocol (IP)
 - It was developed to replace IPv4 due to depletion of IPv4 addresses
 - IPv6 uses 128-bit addresses as compared to 32-bit addresses in IPv4, providing a hugely expanded address space
 - IPv6 aims to simplify aspects of address assignment and network renumbering when compared to IPv4
2. IPv6 address
 - An IPv6 address is 128 bits long, written as 8 groups of 4 hexadecimal digits separated by colons (:)
 - For example, 2001:db8:85a3:0:0:8a2e:370:7334
 - Leading zeros in a group can be omitted, but a single group of consecutive zeros can be abbreviated with a double colon (::)
 - IPv6 addresses are classified into unicast, multicast and anycast addresses
3. IPv6 in China
 - China officially launched IPv6 commercialization in April 2011 and planned to implement IPv6 in three phases by 2025
 - As of June 2018, China has over 120 million IPv6 users which accounts for about 25% of the world’s IPv6 users
 - The fast growth of IPv6 deployment in China is due to government support and efforts by large network operators like China Telecom, China Unicom and China Mobile to promote IPv6 commercialization
 - However, more work is needed to make IPv6 more widely adopted in China to meet the goal of large-scale commercial application by 2025

[Additional points and examples can be added.]

Here is the content in markdown format without any emojis or external links, written in a formal tone with points:

Unit 4 - Transport Layer in Computer Networks

1. Introduction to Transport Layer

- The transport layer is responsible for delivering data between processes executing on different hosts.
- It provides logical communication between application processes running on different network hosts.
- It offers reliable data transmission with error correction and flow control.
- It supports multiplexing allowing multiple applications to use the network simultaneously.
- The two most common transport protocols are TCP and UDP.

2. Transmission Control Protocol (TCP)

- TCP provides reliable, ordered, and error-checked delivery of transmitted data.
- It is a connection-oriented protocol with a three-way handshake to establish a connection between two hosts.
- Data is transmitted in segments with sequence and acknowledgement numbers to ensure ordered delivery.
- Lost or corrupted segments are retransmitted.
- Congestion control is used to avoid overwhelming the network.
- Web transfers, email, and remote login use TCP.

3. User Datagram Protocol (UDP)

- UDP provides fast but unreliable data transmission without error correction or ordering.
- It is a connectionless protocol that does not establish a dedicated end-to-end connection.
- Data is transmitted in datagrams with source and destination port numbers.
- Lost or corrupted datagrams are not retransmitted.
- Streaming media and video conferencing use UDP to prioritize time over reliability.

4. Differences between TCP and UDP

- TCP is reliable, ordered, and error-checked while UDP is fast but unreliable.
- TCP uses a three-way handshake to establish a connection while UDP is connectionless.
- TCP uses sequence and acknowledgement numbers, retransmissions, and congestion control while UDP uses only port numbers and no error correction.
- TCP is typically used for web transfers and email while UDP is used for streaming media and video conferencing.

Here is the formal content on the given topic:

Process-to-process delivery in transport layer

1. The transport layer provides process-to-process delivery of the entire message. It receives messages from the upper layers, splits the messages into segments and reassembles them at the destination.
2. Each segment includes a sequencing number so that the transport layer protocol can reassemble the segments in the correct order at the destination.
3. The transport layer provides an interface between the upper layers and the network layer. It masks the network layer from the upper layers. The transport layer handles segmenting and reassembling the message.
4. The transport layer also controls the flow of segments, including detection of missing and duplicate segments. It provides mechanisms to ensure all segments arrive and are in the correct sequence.
5. Two protocols that provide process-to-process delivery in the transport layer are:
 - Transmission Control Protocol (TCP) which provides a reliable, in-order delivery of bytes.
 - User Datagram Protocol (UDP) which provides fast delivery of unordered messages with no error correction.

The above points cover the key aspects of process-to-process delivery in the transport layer. The transport layer ensures the complete and correct delivery of the messages passed from the upper layers by handling segmentation, reassembly, sequence control and flow control.

Here is the content in markdown format inside header `### Transport layer protocols`:

Transport layer protocols

1. TCP (Transmission Control Protocol):
 - Connection-oriented protocol (establishes dedicated end-to-end connection between two hosts before data transmission)
 - Provides reliable data transmission with in-order delivery
 - Uses 3-way handshake to establish connection
 - Uses sequence and acknowledgement numbers to achieve reliability
 - Provides congestion control to avoid congestion collapse
 - Used by applications requiring high reliability such as HTTP, SMTP, FTP, etc.
2. UDP (User Datagram Protocol):
 - Connectionless protocol (does not establish dedicated end-to-end connection)
 - Provides fast but unreliable data transmission (does not guarantee delivery or ordering)

- Used by applications requiring fast data transmission such as video streaming, DNS, etc. where some loss is tolerable.
3. SCTP (Stream Control Transmission Protocol):
- Connection-oriented protocol
 - Provides reliable and unreliable data transmission
 - Multi-homing and multi-streaming capabilities to provide more robust data transmission
 - Used for signaling and control information in telecommunication networks

The above summary mentions the key characteristics and uses of the major transport layer protocols TCP, UDP and SCTP. The points are written in a formal tone with no emojis or external links as per the given instructions. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the content in Markdown format with formal tone and without external links or emojis:

UDP Transport layer protocol

- UDP stands for User Datagram Protocol. It is a transport layer protocol used for establishing low-latency and loss-tolerating connections between applications on the network.
- UDP is a connectionless protocol as it does not establish a dedicated end-to-end connection between the sending and receiving hosts. It just appends the source and destination port numbers to the header and sends the datagram.
- Due to the lack of connection establishment, the overhead is low and hence UDP is suitable for applications requiring fast transmission of data such as real-time video and audio streaming.
- However, since there is no connection, reliability is not guaranteed. Datagrams may arrive out of order or may not arrive at all. The application using UDP must take care of checking for errors and ensuring complete delivery of data.
- The header size of UDP is 8 bytes, which makes its overhead very less compared to TCP. This makes it a preferred choice for applications with strict latency requirements.
- Some applications that use UDP are DNS, SNMP, Routing protocols, Voice over IP, Video streaming etc.

The above points cover the key highlights of the UDP transport layer protocol in a formal tone with Markdown formatting and without any external links or emojis. Please let me know if you would like me to elaborate on any of the points or modify the content in any way.

Here is the formal content on TCP Transport layer protocol in markdown format:

TCP Transport layer protocol

1. TCP is a transport layer protocol that provides reliable, ordered, and error-checked delivery of transmitted data over IP networks.
2. TCP establishes a dedicated end-to-end connection between two hosts.
3. TCP uses a three-way handshake to establish a connection between two hosts.
4. TCP provides reliable data transmission with the help of sequence and acknowledgement numbers. The sender assigns a sequence number to each byte transmitted and the receiver acknowledges the data by sending an acknowledgement number.
5. TCP provides in-order delivery of data. The receiving host reassembles the received data in the correct order using the sequence numbers.
6. TCP provides flow control and congestion control. The sender cannot overwhelm the receiver with data. TCP uses sliding window protocol for flow control.
7. TCP provides full duplex data transmission. Data can be transmitted in both directions simultaneously.
8. The TCP header includes source and destination port numbers, sequence and acknowledgement numbers, header length, reserved flags, and window size.
9. Common TCP protocols include HTTP, FTP, SMTP, and Telnet. HTTP uses TCP port 80, FTP uses TCP port 21, SMTP uses TCP port 25, and Telnet uses TCP port 23.

Does this sound okay? Let me know if you would like me to modify or add anything.

Here is the formal content on Multiplexing in transport layer in markdown format with points:

Multiplexing in transport layer

1. Multiplexing is the process of combining multiple data streams over a single transmission medium. In the transport layer, multiplexing is used to combine multiple application layer data streams over a single port number. This allows multiple applications to share a single port, conserving system resources.
2. Two types of multiplexing used in the transport layer are:
 - Port number multiplexing: Each application is assigned a port number, allowing the transport layer to direct incoming data to the correct application.
 - Session multiplexing: A session ID is used to associate segments of data with a particular application.
3. Multiplexing is handled differently in TCP and UDP:

- In TCP, port numbers are used to multiplex. Each application must have its own port.
 - In UDP, port numbers are also used but optional. Multiple applications can reuse the same port number if their data can be uniquely identified another way.
4. The benefits of multiplexing include:
- Efficient use of system resources like port numbers and network bandwidth.
 - Allowing multiple applications to run concurrently and share transport layer services.
 - Simplified network architecture. Without multiplexing, each application would require its own transport mechanism.

Does this sound okay? Let me know if you would like me to modify or expand the content in any way.

Here is the content in markdown format with formal tone and without external links or emojis:

Connection management in transport layer

1. Connection establishment: The transport layer protocols establish a connection between the source and destination before the actual transfer of data. This connection ensures that the data is delivered to the correct process on the destination machine.
2. Data transfer: The transport layer transfers the data from the source to the destination over the established connection. It segments and reassembles the data if needed. It also handles any errors that may occur during data transfer.
3. Connection release: Once the transfer of data is complete, the transport layer releases the connection. The resources allocated to the connection are freed for use by other connections.

The key tasks of connection management are:

- Allocating and de-allocating resources for a connection
- Synchronizing the processes on both ends of a connection
- Error recovery by retransmitting data or closing connections

The transport layer protocols, TCP and UDP, provide different types of connections to suit various application requirements. TCP provides a reliable, in-order, byte-stream with congestion control, while UDP provides a fast but unreliable datagram service.

Does this match your requested style? Let me know if you would like me to modify or expand the content in any way.

Here is the content in markdown format with formal tone and without emojis:

Flow control in transport layer

1. Flow control is a technique to regulate the transmission of data to prevent overflow. In transport layer, flow control is implemented to prevent the receiving device from being overwhelmed by too much data sent by the sending device.
2. The receiving device has a fixed buffer size. If data is received faster than it can be processed, the buffer may overflow resulting in loss of data. Flow control mechanisms help avoid this situation by controlling the transmission rate.
3. The most commonly used flow control mechanisms are:
 - Sliding window protocol: The sending device is allowed to send only a fixed number of frames before waiting for an acknowledgement from the receiving device. The window size determines how many frames can be sent at a time.
 - Credit-based flow control: The receiving device sets a limit on the number of bytes that can be received, known as credit. The sending device can only send data if it has credits available. As the receiver processes data, it releases credits back to the sender.
4. Flow control is implemented using feedback messages from the receiver to the sender conveying information about its buffer occupancy. The sender can then adjust its transmission rate accordingly to prevent overflow. Effective flow control mechanisms allow for efficient utilization of the network bandwidth while avoiding congestion.

How's that? I have written the content in points in a formal tone without emojis and external links as instructed. Please let me know if you would like me to modify or add anything.

Here is the content in markdown format with formal tone and without emojis:

Etransmission in Transport Layer

1. Etransmission refers to the process of resending the packets that get corrupted or lost during transmission. It is done by the transport layer to provide reliable data transmission between two hosts.
2. The two transport layer protocols TCP and UDP handle etransmission differently:
 - TCP provides reliable data transmission with etransmission and ordering of packets. It uses sequence numbers and acknowledgments to identify and resend lost or corrupted packets.
 - UDP provides fast but unreliable data transmission without etransmission of packets. It is used for applications that require speed and can tolerate some loss of packets like video streaming.
3. The etransmission process adds some overhead in the form of bandwidth

usage and delay. However, for applications where reliability is important, the benefits of etransmission outweigh these costs. The etransmission process continues until all packets are received correctly at the destination.

4. In summary, etransmission is an important mechanism used by the transport layer to provide reliable data transmission between two hosts. It helps recover from packet loss or corruption for uninterrupted data delivery.

Does this meet your requirements? Let me know if you would like me to modify or expand the content in any way.

Here is the formal content on the given topic in Markdown format:

Window management in transport layer

1. Window size - The amount of data that can be sent before an acknowledgement is received. This is done to improve efficiency and throughput. Eg: If window size is 64 KB, sender can send 64 KB of data before waiting for acknowledgement.
2. Sliding window - Window size 'slides' over the data as acknowledgements are received. This helps in continuous transmission of data. eg: If first 64 KB is sent and acknowledgement received, next 64 KB is sent. This 'slides' the window over data.
3. Window full - If acknowledgements are not received in time, window gets full and sender stops sending data. This avoids overflow and congestion.
4. Advertised window - Receiver advertises its available buffer space as window size to inform sender how much data it can accommodate. This helps sender send data accordingly and avoid overflow at receiver.

The above points cover the key aspects of window management in transport layer which controls the amount of data in transit and improves throughput. Window size, sliding window, window full situation and advertised window are the main concepts in this method of efficient data transmission.